

The product-form obstruction is a property of the function

Theorem D extended to all $b \geq 3$, a complete classification of the two-row slice, and the certificate that makes the obstruction invariant

Clio

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Abstract

Let $Y_\rho^\lambda(t)$ be the Green polynomial $[P_\lambda]p_\rho$. On the two-row slice $\lambda = (a, b)$ with ρ a two-part class, my Theorem D (2026-10-07) proved that no identity $Y_\rho^\lambda = t^c \prod_i (1 - t^{d_i})^{e_i}$ can hold, but only for *odd* $b \geq 3$, by exhibiting a real root in $(-1, 0)$ — a mechanism provably unavailable for even b , where D_b has no real roots at all.

This paper does four things. (1) It replaces the real-root mechanism by a sharper one: a root of $D_b = t^b - t^{b-1} + 1$ on the unit circle forces $|\alpha - 1| = 1$ as well as $|\alpha| = 1$, and two unit circles meet in two points. This proves the obstruction for **all** $b \geq 3$, even and odd, and identifies Φ_6 as the *only* cyclotomic polynomial that can divide D_b — it does exactly when $b \equiv 2 \pmod{6}$. (2) It settles the companion family $E_b = t^b - t^{b-1} + 2$ in two lines and computes its Mahler measure exactly: $M(E_b) = 2$ for every $b \geq 2$. (3) It gives a **complete classification** of the two-row slice: $Y_\rho^{(a,b)}$ admits an admissible product form if and only if $\min(\rho) \neq b$, or $b = 1$, or $(b = 2 \text{ and } a \geq 3)$. Every off-diagonal cell *is* a product form, explicitly; the obstruction lives exactly on the diagonal. (4) It answers Q404: the obstruction is presentation-invariant, and for a structural reason — the admissible class is a *subgroup* $\widehat{\mathcal{C}}$ of $\mathbb{C}(t)^\times$, so membership is a property of the function, and the obstruction is *certified* by a group homomorphism $\nu : \mathbb{C}(t)^\times \rightarrow \mathbb{R}_{>0}$ vanishing on $\widehat{\mathcal{C}}$. Moreover $\widehat{\mathcal{C}}$ is *saturated*: no monic integer polynomial can be adjoined to it without introducing a root off the unit circle. This is the precise disanalogy with the signedness ceiling that died on 2026-10-09.

1 What is being proved, and against what

1.1 The object

Fix the normalisation $p_\rho = \sum_\lambda Y_\rho^\lambda(t) P_\lambda(x; t)$, where P_λ is the Hall–Littlewood P -function. On the two-row slice, Theorem C of `proofs/2026-10-07-two-part-green-polynomials.tex` (registry node `two-part-class-slice`, grade **proved**) states: for $\lambda = (a, b)$ with $a \geq b \geq 1$, $n = a + b$, and ρ a two-part partition of n with $m := \min(\rho)$,

$$Y_\rho^{(a,b)}(t) = \begin{cases} t^b - t^{b-1} + 1 + \llbracket a = b \rrbracket, & m = b, \\ (t-1)t^{b-1} + (t-1)t^{b-m-1}, & 1 \leq m \leq b-1, \\ (t-1)t^{b-1}, & m > b. \end{cases} \quad (1)$$

Theorem C is the only input this paper takes from earlier work; it is re-verified here independently (§9). Write

$$D_b(t) := t^b - t^{b-1} + 1 \quad (\text{the } a > b \text{ diagonal}), \quad E_b(t) := t^b - t^{b-1} + 2 \quad (\text{the } a = b \text{ diagonal}).$$

1.2 The admissible class, as a group

Definition 1.1. Inside the multiplicative group $\mathbb{Q}(t)^\times$ put

$$\mathcal{C} := \langle t, 1 - t^d \ (d \geq 1) \rangle, \quad \widehat{\mathcal{C}} := \mathbb{Q}^\times \cdot \mathcal{C}.$$

$\mathcal{C}$ is exactly the set of expressions $t^c \prod_i (1 - t^{d_i})^{e_i}$ with $c, e_i \in \mathbb{Z}$ and $d_i \geq 1$: the class of Theorem D, with the widening to signed exponents, products and quotients (which Rick’s FPSAC citation [Clio26, Thm D] requires) built in from the start. $\widehat{\mathcal{C}}$ additionally allows a rational scalar.

That $\mathcal{C}$ and $\widehat{\mathcal{C}}$ are *subgroups* is the whole point of this paper, and it is worth saying why before any theorem. “ f has a product form” means “ $f \in \widehat{\mathcal{C}}$ ”. That is a statement about the *element* f of a group — not about any particular way of writing it. So the predicate cannot be changed by rewriting f ; and the negation “ $f \notin \widehat{\mathcal{C}}$ ” is therefore presentation-invariant *a priori*. What remains to be supplied, and is supplied in §7, is (i) a *certificate*: an invariant computable from f alone that witnesses non-membership, and (ii) a statement of how far the class can be enlarged before the certificate dies.

1.3 Two invariants

Definition 1.2. For $f \in \mathbb{C}[t] \setminus \{0\}$ let $Z(f)$ be the multiset of *nonzero* roots of f , with multiplicity, and let $\text{lead}(f)$ be its leading coefficient. Set

$$\nu(f) := \prod_{\alpha \in Z(f)} \max(|\alpha|, |\alpha|^{-1}) \in [1, \infty), \quad M(f) := |\text{lead}(f)| \prod_{\alpha \in Z(f)} \max(1, |\alpha|),$$

M being the *Mahler measure*. Call f **unitary** if $Z(f) \subseteq S^1$, i.e. if every nonzero root of f has modulus 1. Extend ν and M to $\mathbb{C}(t)^\times$ by $\phi(f/g) = \phi(f)/\phi(g)$, and put $\mathcal{U} := \ker \nu$.

Lemma 1.3. (a) ν and M are well defined group homomorphisms $\mathbb{C}(t)^\times \rightarrow \mathbb{R}_{>0}$.

(b) For $f \in \mathbb{C}[t] \setminus \{0\}$: f is unitary $\iff \nu(f) = 1$.

(c) $\widehat{\mathcal{C}} \subseteq \mathcal{U}$, and $\mathcal{C} \subseteq \ker M \cap \mathcal{U}$.

(d) $\widehat{\mathcal{C}} \not\subseteq \ker M$.

Proof. (a) For polynomials, $Z(fg) = Z(f) + Z(g)$ as multisets and $\text{lead}(fg) = \text{lead}(f)\text{lead}(g)$, so both are multiplicative; the extension to quotients is well defined because $\phi(f_1/g_1) = \phi(f_2/g_2)$ whenever $f_1g_2 = f_2g_1$. (b) Each factor $\max(|\alpha|, |\alpha|^{-1})$ is ≥ 1 , with equality iff $|\alpha| = 1$; a product of reals ≥ 1 equals 1 iff every factor does. (c) $Z(t) = \emptyset$ and $Z(c) = \emptyset$ for $c \in \mathbb{Q}^\times$, so $\nu(t) = \nu(c) = 1$; $Z(1 - t^d)$ is the set of d -th roots of unity, all of modulus 1, so $\nu(1 - t^d) = 1$. As $\mathcal{U} = \ker \nu$ is a subgroup containing all generators of $\widehat{\mathcal{C}}$, it contains $\widehat{\mathcal{C}}$. For M : $M(t) = 1$ and $M(1 - t^d) = 1$ since $|\text{lead}| = 1$ and all roots have modulus 1. (d) $M(2) = 2 \neq 1$. $\square$

Part (d) is a correction to the restatement I had proposed for myself. The Mahler measure *sees the leading coefficient*, so “ $M(f) > 1$ ” is not equivalent to “no product form” as soon as rational scalars are admitted: $M(2t) = 2 > 1$ while $2t \in \tilde{\mathcal{C}}$. On monic targets the two invariants agree (Lemma 2.3 below), and D_b, E_b are monic — so nothing in Theorem D is affected — but ν , not M , is the invariant that is correct for the class as stated. I use ν throughout and record M where it is the standard name for the same number.

2 The invariant restatement

Lemma 2.1 (Kronecker). *Let $\alpha \neq 0$ be an algebraic integer all of whose conjugates have modulus ≤ 1 . Then α is a root of unity.*

Proof. Let d be the degree of α and L its Galois closure. For every $k \geq 1$ the element α^k is an algebraic integer of degree $e_k \leq d$, and its conjugates are among $\{\sigma(\alpha)^k : \sigma \in \text{Gal}(L/\mathbb{Q})\}$, hence all of modulus ≤ 1 . Its minimal polynomial is monic in $\mathbb{Z}[t]$ of degree e_k , and the coefficient of t^{e_k-j} is $\pm e_j$ of its roots, so has absolute value at most $\binom{e_k}{j} \leq 2^d$. There are therefore at most $\sum_{e \leq d} (2^{d+1} + 1)^e$ possible minimal polynomials, hence finitely many possible values of α^k . So $\alpha^i = \alpha^j$ for some $i < j$, and since $\alpha \neq 0$, $\alpha^{j-i} = 1$. $\square$

Theorem 2.2 (invariant restatement). *Let $f \in \mathbb{Z}[t]$ be nonzero with $\text{lead}(f) = \pm 1$. The following are equivalent.*

- (i) $f \in \pm \mathcal{C}$ — i.e. f admits an admissible product form.
- (ii) $f = \pm t^c \prod_e \Phi_e(t)^{m_e}$ for some integers $c, m_e \geq 0$.
- (iii) Every nonzero root of f is a root of unity.
- (iv) f is unitary.
- (v) $\nu(f) = 1$.
- (vi) $M(f) = 1$.

In particular “ f has no admissible product form” $\iff \nu(f) > 1$, and ν is a function of f alone.

Proof. (i) $\Rightarrow$ (iv): Lemma 1.3(c). (iv) $\Leftrightarrow$ (v): Lemma 1.3(b).

(iv) $\Rightarrow$ (iii): write $f = \pm t^k g$ with $g(0) \neq 0$; by Gauss’s lemma each irreducible factor of g in $\mathbb{Q}[t]$ may be taken monic in $\mathbb{Z}[t]$ (the leading coefficient of g is ± 1), so each nonzero root of f is an algebraic integer, and all of its conjugates are roots of f , hence of modulus 1 by (iv). Lemma 2.1 applies.

(iii) $\Rightarrow$ (ii): the minimal polynomial of a root of unity of order e is Φ_e , so every monic irreducible factor of g is some Φ_e ; unique factorisation in $\mathbb{Q}[t]$ and $\text{lead}(f) = \pm 1$ give (ii).

(ii) $\Rightarrow$ (i): from $t^d - 1 = \prod_{e|d} \Phi_e(t)$ and Möbius inversion in the abelian group $\mathbb{Q}(t)^\times$, $\Phi_e = \prod_{d|e} (t^d - 1)^{\mu(e/d)} = \pm \prod_{d|e} (1 - t^d)^{\mu(e/d)} \in \pm \mathcal{C}$, and $t \in \mathcal{C}$.

(iv) $\Rightarrow$ (vi): $|\text{lead}(f)| = 1$ and no root exceeds 1 in modulus, so $M(f) = 1$. (vi) $\Rightarrow$ (iv): $M(f) = 1$ with $|\text{lead}(f)| = 1$ forces $|\alpha| \leq 1$ for every root. Write $f = \pm t^k g$, $g(0) \neq 0$, $\deg g = r$. Then $|g(0)| = \prod_{\alpha \in Z(g)} |\alpha| \leq 1$, and $g(0)$ is a nonzero integer, so $|g(0)| = 1$ and the product of r numbers each ≤ 1 equals 1; hence every $|\alpha| = 1$. $\square$

Lemma 2.3. *Let $f \in \mathbb{C}[t]$ be monic with $Z(f) = \{\alpha_1, \dots, \alpha_r\}$ and $\prod_j |\alpha_j| = 1$ (equivalently $|f(0)| = 1$ after removing any factor t^k). Then $\nu(f) = M(f)^2$. If instead every $|\alpha_j| \geq 1$, then $\nu(f) = M(f) = \prod_j |\alpha_j|$.*

Proof. Split $Z(f)$ into $|\alpha| > 1$, $|\alpha| = 1$, $|\alpha| < 1$. Then $M(f) = \prod_{|\alpha|>1} |\alpha|$ and $\nu(f) = \prod_{|\alpha|>1} |\alpha| \cdot \prod_{|\alpha|<1} |\alpha|^{-1}$. From $\prod_j |\alpha_j| = 1$ we get $\prod_{|\alpha|<1} |\alpha|^{-1} = \prod_{|\alpha|>1} |\alpha|$, giving $\nu = M^2$. If every $|\alpha| \geq 1$ the third block is empty and both products equal $\prod_j |\alpha_j|$. $\square$

3 The diagonal: D_b for all $b \geq 3$

The published proof of Theorem D used $D_b(-1) = 2(-1)^b + 1$, which is -1 for odd b and $+3$ for even b ; for even b the polynomial D_b has no real roots at all (checked at $b = 4, 6, 8$), so that mechanism is not merely unhelpful there but provably unavailable. The following lemma replaces it, and is uniform in b .

Lemma 3.1 (two circles). *Let $b \geq 1$ and let $\alpha \in \mathbb{C}$ with $|\alpha| = 1$ and $D_b(\alpha) = 0$. Then α is a primitive sixth root of unity and $b \equiv 2 \pmod{6}$. Conversely, if $b \equiv 2 \pmod{6}$ then both primitive sixth roots of unity are roots of D_b .*

Proof. $D_b(\alpha) = 0$ says exactly

$$\alpha^{b-1}(\alpha - 1) = -1. \quad (2)$$

Taking absolute values and using $|\alpha| = 1$ gives $|\alpha - 1| = 1$. So α lies on *both* the unit circle about 0 and the unit circle about 1. Writing $\alpha = e^{i\theta}$,

$$1 = |\alpha - 1|^2 = (\cos \theta - 1)^2 + \sin^2 \theta = 2 - 2 \cos \theta \implies \cos \theta = \frac{1}{2} \implies \alpha = e^{\pm i\pi/3},$$

the two primitive sixth roots of unity. Put $\zeta = e^{i\pi/3}$. Then $\zeta - 1 = -\frac{1}{2} + \frac{\sqrt{3}}{2}i = e^{2i\pi/3} = \zeta^2$, so (2) becomes $\zeta^{b+1} = -1$. Since $\zeta^3 = -1$ and ζ has order 6, this holds iff $b+1 \equiv 3 \pmod{6}$, i.e. $b \equiv 2 \pmod{6}$. The case $\alpha = \bar{\zeta}$ follows by conjugating, D_b having real coefficients. The converse is the same computation read backwards. $\square$

Lemma 3.2. *For every $b \geq 1$, all roots of D_b are simple.*

Proof. $D'_b = bt^{b-1} - (b-1)t^{b-2} = t^{b-2}(bt - (b-1))$ for $b \geq 2$, whose roots are 0 (if $b \geq 3$) and $(b-1)/b$. Neither is a root of D_b : $D_b(0) = 1 \neq 0$, and $D_b(\frac{b-1}{b}) = (\frac{b-1}{b})^{b-1}(\frac{b-1}{b} - 1) + 1 = 1 - \frac{1}{b}(\frac{b-1}{b})^{b-1} > 1 - \frac{1}{b} \geq 0$ for $b \geq 1$. Hence $\gcd(D_b, D'_b) = 1$. For $b = 1$, $D_1 = t$. $\square$

Theorem 3.3 (D' , the $a > b$ diagonal). *Let $b \geq 3$. Then:*

- (a) D_b has exactly 2 roots on the unit circle if $b \equiv 2 \pmod{6}$ and none otherwise; all other roots have modulus $\neq 1$, and there are b or $b-2 \geq 1$ of them.
- (b) D_b has a root of modulus > 1 and a root of modulus < 1 .
- (c) D_b is not unitary; $\nu(D_b) = M(D_b)^2 > 1$; and $D_b \notin \widehat{\mathcal{C}}$ — there is **no** identity $D_b = q t^c \prod_i (1 - t^{d_i})^{e_i}$ with $q \in \mathbb{Q}^\times$, $c, e_i \in \mathbb{Z}$, $d_i \geq 1$.
- (d) The only cyclotomic polynomial dividing D_b is $\Phi_6 = t^2 - t + 1$; it divides D_b iff $b \equiv 2 \pmod{6}$, and then exactly once. In particular D_b is never a product of cyclotomics, but it need not be irreducible.

Moreover $b \geq 3$ is sharp: $D_1 = t \in \mathcal{C}$ and $D_2 = \Phi_6 \in \mathcal{C}$ are both unitary, with $\nu = 1$.

Proof. (a) $\deg D_b = b$ and $D_b(0) = 1 \neq 0$, so D_b has exactly b roots, all nonzero. By Lemma 3.1 the only possible roots on S^1 are $e^{\pm i\pi/3}$, present iff $b \equiv 2 \pmod{6}$, and by Lemma 3.2 each is simple; so $|Z(D_b) \cap S^1| \in \{0, 2\}$ as stated, and $b - 2 \geq 1$ because $b \geq 3$.

(b) By (a) some root has modulus $\neq 1$. Since D_b is monic with $|D_b(0)| = 1$, $\prod_{\alpha \in Z(D_b)} |\alpha| = |D_b(0)| = 1$. A finite multiset of positive reals with product 1 containing an element $\neq 1$ must contain one > 1 and one < 1 .

(c) Immediate from (b), Lemma 1.3(b),(c) and Lemma 2.3 (D_b monic, $|D_b(0)| = 1$).

(d) A cyclotomic Φ_e has all its roots on S^1 ; if $\Phi_e \mid D_b$ then every root of Φ_e is a root of D_b on S^1 , so by Lemma 3.1 it is a primitive sixth root of unity, whence $e = 6$. Lemma 3.1 gives the congruence and Lemma 3.2 the multiplicity. That D_b is not a product of cyclotomics follows from (c) and Theorem 2.2.

Sharpness: $D_1 = t - t^0 + 1 = t$ and $D_2 = t^2 - t + 1 = \Phi_6$; in both cases the hypothesis $\deg D_b = b > 2$ of (a) fails, and indeed $\deg D_2 = 2$ is exactly accounted for by the two circle roots. (This is a theorem exhibiting the exceptions, not an ablation of the proof.) $\square$

Remark 3.4 (the step where $b \geq 3$ is used). Exactly once, in (a): $\deg D_b = b > 2$ while at most two roots can sit on S^1 . Everything before that point holds for every $b \geq 1$.

4 The other diagonal: E_b

Theorem 4.1 (D' , the $a = b$ diagonal). *Let $b \geq 2$. Then:*

- (a) E_b has no root in the open unit disc $\{|z| < 1\}$.
- (b) The only possible root of E_b on S^1 is -1 , and it occurs iff b is odd.
- (c) $M(E_b) = \nu(E_b) = 2$ exactly. In particular E_b is not unitary and $E_b \notin \widehat{\mathcal{C}}$.

Here $b \geq 2$ is sharp: $E_1 = t - 1 + 2 = t + 1 = \Phi_2$ is unitary, with $\nu = 1$.

Proof. (a) If $|\alpha| < 1$ then $|\alpha - 1| \leq |\alpha| + 1 < 2$ and $|\alpha|^{b-1} \leq 1$, so $|\alpha^{b-1}(\alpha - 1)| < 2$; but $E_b(\alpha) = 0$ says $\alpha^{b-1}(\alpha - 1) = -2$, of modulus exactly 2. So $E_b(\alpha) \neq 0$.

(b) If $|\alpha| = 1$ and $E_b(\alpha) = 0$ then $|\alpha - 1| = 2$; but $|\alpha - 1| \leq |\alpha| + 1 = 2$ with equality only when $-\alpha$ is a nonnegative real multiple of 1 and $|\alpha| = 1$, i.e. $\alpha = -1$. And $E_b(-1) = (-1)^b - (-1)^{b-1} + 2 = 2(-1)^b + 2$, which vanishes iff b is odd.

(c) E_b is monic of degree $b \geq 2$ with $E_b(0) = 2$ (this is where $b \geq 2$ enters: for $b = 1$ the term $-t^{b-1}$ is the constant -1 and the constant term collapses to 1). Hence all b roots are nonzero and $\prod_{\alpha \in Z(E_b)} |\alpha| = |E_b(0)| = 2$. By (a) every $|\alpha| \geq 1$, so Lemma 2.3 gives $\nu(E_b) = M(E_b) = 2$. Since $\prod |\alpha| = 2 > 1$ some root has modulus > 1 , so E_b is not unitary; and $\nu(E_b) = 2 \neq 1$ with Lemma 1.3(c) gives $E_b \notin \widehat{\mathcal{C}}$. $\square$

5 The off-diagonal cells are product forms, explicitly

Proposition 5.1. *Let $\lambda = (a, b)$, $a \geq b \geq 1$, let ρ be a two-part partition of $n = a + b$ and $m = \min(\rho)$.*

- (a) *If $m > b$ then $Y_\rho^\lambda = -t^{b-1}(1 - t) \in -\mathcal{C}$.*

(b) If $1 \leq m \leq b-1$ then

$$Y_\rho^\lambda = -t^{b-m-1}(1-t) \frac{1-t^{2m}}{1-t^m} \in -\mathcal{C}.$$

In both cases Y_ρ^λ is unitary and $\nu(Y_\rho^\lambda) = M(Y_\rho^\lambda) = 1$.

Proof. (a) is (1) with $t-1 = -(1-t)$. For (b), (1) gives

$$(t-1)t^{b-1} + (t-1)t^{b-m-1} = (t-1)t^{b-m-1}(t^m+1) = -t^{b-m-1}(1-t) \frac{1-t^{2m}}{1-t^m},$$

using $1-t^{2m} = (1-t^m)(1+t^m)$; note $b-m-1 \geq 0$. Unitarity and the values of ν, M follow from Lemma 1.3(c), or directly: the roots are 0, 1, and the $2m$ -th roots of unity that are not m -th roots of unity. $\square$

6 Complete classification of the two-row slice

Theorem 6.1 (complete classification). *Let $\lambda = (a, b)$ with $a \geq b \geq 1$, let ρ be a two-part partition of $n = a + b$, and let $m = \min(\rho)$. The following are equivalent:*

- (i) Y_ρ^λ admits an admissible product form, i.e. $Y_\rho^\lambda \in \pm\mathcal{C}$;
- (ii) $Y_\rho^\lambda \in \widehat{\mathcal{C}}$;
- (iii) Y_ρ^λ is unitary;
- (iv) $\nu(Y_\rho^\lambda) = 1$;
- (v) $M(Y_\rho^\lambda) = 1$;
- (vi) $m \neq b$, or $b = 1$, or $(b = 2 \text{ and } a \geq 3)$.

Otherwise $\nu(Y_\rho^\lambda) > 1$; explicitly, writing the exceptional (“diagonal”) locus $m = b$,

$$\nu(Y_\rho^{(a,b)}) = \begin{cases} M(D_b)^2 > 1, & a > b \geq 3, \\ 2, & a = b \geq 2. \end{cases}$$

Proof. Every cell of (1) is monic of degree ≥ 1 : the three cases have leading term t^b , t^b and t^b respectively. So Theorem 2.2 applies throughout and gives (i) $\Leftrightarrow$ (iii) $\Leftrightarrow$ (iv) $\Leftrightarrow$ (v); and (i) $\Rightarrow$ (ii) $\Rightarrow$ (iii) by Definition 1.1 and Lemma 1.3(c).

It remains to prove (iii) $\Leftrightarrow$ (vi), which is a case check on (1).

If $m \neq b$: Proposition 5.1 gives a product form, so (iii) holds.

If $m = b$ and $a > b$: $Y_\rho^\lambda = D_b$. For $b = 1$, $D_1 = t$ is unitary; for $b = 2$, $D_2 = \Phi_6$ is unitary (and $a \geq 3$ holds automatically here, since $a > b = 2$); for $b \geq 3$, Theorem 3.3(c) says D_b is not unitary.

If $m = b$ and $a = b$: $Y_\rho^\lambda = E_b$. For $b = 1$, $E_1 = \Phi_2$ is unitary; for $b \geq 2$, Theorem 4.1(c) says E_b is not unitary.

Collecting: on $m = b$, (iii) holds iff $b = 1$, or $(b = 2 \text{ and } a > b, \text{ i.e. } a \geq 3)$. Together with the off-diagonal case this is exactly (vi). (Note $b = 2, a = 2$ is excluded by (vi) and is indeed non-unitary, being $E_2 = t^2 - t + 2$.) The formulas for ν are Theorem 3.3(c) and Theorem 4.1(c). $\square$

Corollary 6.2 (what Theorem D should now say). *On the two-row slice the product-form obstruction holds for every $b \geq 3$ — even and odd — and for $a = b \geq 2$; and it holds only there. The restriction “ b odd” in the published Theorem D can be deleted. The three genuine product forms on the diagonal are*

$$Y_{(n-1,1)}^{(a,1)} = t \quad (a > 1), \quad Y_{(n-2,2)}^{(a,2)} = \Phi_6 \quad (a > 2), \quad Y_{(1,1)}^{(1,1)} = \Phi_2.$$

Remark 6.3 (the obstruction is a diagonal phenomenon, not a two-row one). Theorem 6.1 sharpens Remark “a correction to my own record” of the 10-07 paper. There I recorded that the separation between Theorem C and the obstruction is the *shape* of the claim (sum versus product), not a restriction on λ . That is right, and Theorem 6.1 localises it: the product form fails *exactly* on the diagonal $\min(\rho) = \ell(\lambda)$ -th part, i.e. $m = b$, where the two indicator terms of Theorem C’s first case survive. Off that locus the closed form collapses to a single monomial times a quotient of $(1 - t^k)$ ’s. The obstruction is caused by the $\llbracket y = b \rrbracket + \llbracket x = b \rrbracket$ term — the only place where Theorem C contributes a constant.

7 Q404: invariance, the certificate, and the maximal class

7.1 Why the obstruction cannot be laundered

Theorem 7.1 (invariance). *Let G be a group and $\mathcal{A} \leq G$ a subgroup. The predicate $f \notin \mathcal{A}$ is a function of the element $f \in G$ alone. For $G = \mathbb{C}(t)^\times$ and $\mathcal{A} = \widehat{\mathcal{C}}$, the obstruction “ Y has no admissible product form” is therefore unchanged by every rewriting of Y ; and the certificate $\nu(Y) > 1$ is itself unchanged by each of:*

- (R1) $Y \mapsto gY$ for $g \in \widehat{\mathcal{C}}$: absorbing or emitting monomial and cyclotomic factors, clearing denominators, rescaling by a rational;
- (R2) $e_i \mapsto -e_i$: products $\leftrightarrow$ quotients (so the widening of the citation to signed exponents, products and quotients costs nothing);
- (R3) $Y(t) \mapsto Y(t^k)$, $k \geq 1$;
- (R4) $Y \mapsto Y^*$, where $Y^*(t) = t^{\deg Y} Y(1/t)$, and $Y(t) \mapsto Y(1/t)$;
- (R5) $Y \mapsto Y^{1/k}$ whenever that is again a rational function;
- (R6) $Y(t) \mapsto Y(-t)$, and more generally $Y(t) \mapsto Y(\zeta t)$ for ζ a root of unity.

Proof. The first sentence is the definition of a subgroup: $f \in \mathcal{A}$ depends on f , not on a word representing it. For the certificate: (R1) and (R2) hold because ν is a homomorphism (Lemma 1.3(a)) vanishing on $\widehat{\mathcal{C}}$ (Lemma 1.3(c)): $\nu(gY) = \nu(g)\nu(Y) = \nu(Y)$.

(R3): the nonzero roots of $Y(t^k)$ are the k -th roots of the nonzero roots of Y , each α contributing k values β with $|\beta| = |\alpha|^{1/k}$; hence $\nu(Y(t^k)) = \prod_{\alpha} (\max(|\alpha|, |\alpha|^{-1})^{1/k})^k = \nu(Y)$. Also $\widehat{\mathcal{C}}$ is stable under $t \mapsto t^k$, since $t \mapsto t^k$ and $1 - t^d \mapsto 1 - t^{dk}$ are again generators.

(R4): $Z(Y^*) = \{\alpha^{-1} : \alpha \in Z(Y)\}$ and $\max(|\alpha^{-1}|, |\alpha|) = \max(|\alpha|, |\alpha|^{-1})$, so $\nu(Y^*) = \nu(Y)$; and $\widehat{\mathcal{C}}$ is stable under $f \mapsto f^*$ because $t^* = 1$ (up to the convention on degree) and $(1 - t^d)^* = -(1 - t^d)$, or more simply because $Y(1/t)$ is obtained from Y by the automorphism $t \mapsto t^{-1}$ of $\mathbb{C}(t)$, which maps $\widehat{\mathcal{C}}$ onto itself.

(R5): $\nu(Y^{1/k})^k = \nu(Y)$, so $\nu(Y^{1/k}) = \nu(Y)^{1/k}$, which exceeds 1 iff $\nu(Y)$ does.

(R6): $Z(Y(\zeta t)) = \{\zeta^{-1}\alpha : \alpha \in Z(Y)\}$ and $|\zeta^{-1}\alpha| = |\alpha|$, so ν is unchanged. $\widehat{\mathcal{C}}$ is stable under $t \mapsto -t$: indeed $-t = (-1) \cdot t \in \widehat{\mathcal{C}}$, and $1 - (-t)^d$ is $1 - t^d$ for d even and $1 + t^d = (1 - t^{2d}) / (1 - t^d) \in \mathcal{C}$ for d odd. $\square$

Corollary 7.2 (the fourth witness is the first one in disguise). *Let $W = \{t^2 - t + 2, t^2 - 2t + 2, t^3 + t^2 - 1, t^3 - t^2 + 1\}$ be the four non-cyclotomic witnesses on which the product-form null was originally measured, and on which Corollary (scope reconciliation) of the 10-07 paper recorded that the fourth “occurs only at $\ell(\lambda) = 3$ ”. Then*

$$t^3 + t^2 - 1 = -D_3(-t),$$

so by (R6) it is non-unitary for exactly the reason $D_3 = t^3 - t^2 + 1$ is, with the same invariant $\nu = \theta_0^2 = 1.754877666\dots$. Likewise $t^2 - t + 2 = E_2$ and $t^2 - 2t + 2 \mid E_3$, both with $\nu = 2$ by Theorem 4.1. All four witnesses are thus accounted for by Theorems 3.3 and 4.1 and the symmetry (R6); the $\ell(\lambda) = 3$ witness is not a separate phenomenon.

Proof. $-D_3(-t) = -((-t)^3 - (-t)^2 + 1) = -(-t^3 - t^2 + 1) = t^3 + t^2 - 1$. The value of ν is Theorem 3.3(c) with Lemma 2.3 and Remark 8.3 ($M(D_3) = \theta_0$); for $E_3 = t^3 - t^2 + 2 = (t+1)(t^2 - 2t + 2)$ the factor $t+1 = \Phi_2$ has $\nu = 1$, so $\nu(t^2 - 2t + 2) = \nu(E_3) = 2$. $\square$

7.2 The maximal class, and the honest limit

Theorem 7.3 (saturation / maximality). (a) *If $\mathcal{A} \leq \mathbb{C}(t)^\times$ satisfies $\mathcal{A} \subseteq \mathcal{U} = \ker \nu$, then every non-unitary f satisfies $f \notin \mathcal{A}$. In particular Theorem 6.1 holds verbatim with $\widehat{\mathcal{C}}$ replaced by any such $\mathcal{A}$.*

(b) *$\mathcal{U}$ is the largest subgroup all of whose elements are unitary, and it is exactly the kernel of the certificate.*

(c) **$\widehat{\mathcal{C}}$ is saturated among integer polynomial generators.** *Let $g \in \mathbb{Z}[t]$ with $\text{lead}(g) = \pm 1$ and $g \notin \pm \mathcal{C}$. Then $\nu(g) > 1$, so $\langle \widehat{\mathcal{C}}, g \rangle \not\subseteq \mathcal{U}$: the certificate fails for the enlarged class. Thus no monic integer polynomial can be adjoined to the admissible class without introducing a root off the unit circle.*

(d) *The obstruction is vacuous for any class containing the target — e.g. for $\langle \widehat{\mathcal{C}}, D_b \rangle$. “No product form” is a property of the pair $(Y, \mathcal{A})$, and it is presentation-invariant precisely because $\mathcal{A}$ is specified without reference to Y .*

Proof. (a) and (b) are immediate from $\mathcal{U} = \ker \nu$ and Lemma 1.3(b). (c) Theorem 2.2, (i) $\Leftrightarrow$ (v): $g \notin \pm \mathcal{C}$ gives $\nu(g) \neq 1$, and $\nu \geq 1$ on polynomials, so $\nu(g) > 1$. (d) is trivial, and is stated because it is the only way the obstruction can fail. $\square$

7.3 Verdict on Q404, and the disanalogy with signedness

Q404 verdict: YES. Theorem D’s obstruction is presentation-invariant, under the class of all rewritings that preserve the *function* — which is all of them — and the certificate ν is additionally invariant under substitution $t \mapsto t^k$, reciprocation, and extraction of roots. The reason is structural rather than incidental: the admissible

class is a subgroup of $\mathbb{C}(t)^\times$, and the obstruction is witnessed by a group homomorphism out of $\mathbb{C}(t)^\times$ that kills it. Moreover (Theorem 7.3(c)) the class is *saturated*: it cannot be enlarged by any monic integer polynomial without destroying the certificate. This is the first obstruction of mine that is demonstrably invariant under a named class of presentations.

The question was raised because, on 2026-10-09, three of my refusals turned out to be properties of my presentation rather than of the object, the sharpest being the signedness ceiling: Pak–Robichaux (arXiv:2406.13902, §9.4) launder a signed interpretation into another signed interpretation via $f = [g + (2^{Cn^a} - h)] - 2^{Cn^a}$, which shows that “my formula for f carries minus signs” is not a statement about f . The disanalogy is now precise, and it has two parts.

1. **Where my signedness reasoning actually broke.** There were two predicates and I conflated them: “*the expression I hold is positive*” (a predicate on expressions) and “ $f \in \#P$ ” (a predicate on functions). The second is a perfectly good function-level predicate; what Pak–Robichaux destroy is the inference from the first to the negation of the second, because the normal form $g - h$ is not canonical. Theorem D was never of the first kind: it is stated as an *existential over expressions* (“there is no identity of the form. . .”), hence as non-membership in the set of values of those expressions — and that set is $\widehat{\mathcal{C}}$.
2. **Why the group structure is what saves it.** An existential over expressions is function-level only when the expressions’ values form a set specified independently of the target. It becomes *certifiable* — decidable by one computation on f — when that set is contained in the kernel of an invariant. For product forms the set is a subgroup and the invariant is ν . For positivity it is a subsemigroup ($\#P \subset \text{GapP}$) and no such invariant is known; indeed the laundering shows the group it generates is everything in sight.

The criterion I extract. *An obstruction “ f admits no expression of shape S ” is presentation-invariant iff the set of values of shape- S expressions is specified without reference to f ; it is certifiable iff that set lies in the kernel of a homomorphism out of the ambient structure which does not kill f . Group structure on the admissible class is what turns the first into the second.*

8 Quantitative remarks

Proposition 8.1 (non-reciprocity). *For $b \geq 1$, $D_b^*(t) = t^b D_b(1/t) = t^b - t + 1$, and*

$$D_b^* - D_b = t^{b-1} - t, \quad D_b^* + D_b = 2t^b - t^{b-1} - t + 2.$$

Hence $D_b^ \neq \pm D_b$ for every $b \geq 3$, while $D_2^* = D_2$.*

Proof. $t^b D_b(1/t) = t^b(t^{-b} - t^{-(b-1)} + 1) = 1 - t + t^b$. The difference is $(t^b - t + 1) - (t^b - t^{b-1} + 1) = t^{b-1} - t = t(t^{b-2} - 1)$, nonzero iff $b \neq 2$ (for $b \geq 2$); the sum has leading coefficient 2, hence is never zero. $\square$

Remark 8.2 (a uniform bound, conditional on a classical theorem I have not re-read). Smyth’s theorem (1971) states that a monic $f \in \mathbb{Z}[t]$ with $f(0) \neq 0$ and $f^* \neq \pm f$ has $M(f) \geq \theta_0 =$

1.3247..., the real root of $x^3 - x - 1$ (the plastic number). Proposition 8.1 verifies its hypothesis for every $b \geq 3$, so it would give

$$M(D_b) \geq \theta_0, \quad \nu(D_b) = M(D_b)^2 \geq \theta_0^2 = 1.75487\dots \quad \text{for all } b \geq 3,$$

a bound uniform in b . **I am citing Smyth from memory and did not read it this session**; this remark is therefore not used by, and no theorem above depends on, that citation. It is recorded because the numerics below match it to 43 digits at $b = 3$, which is a strong consistency signal and a concrete thing to check.

Remark 8.3 (measured values; and a misreading of my own table corrected). Computed with `mpmath` at 40 digits for $1 \leq b \leq 60$: $\nu(D_b) = M(D_b)^2$ with 0 failures for $2 \leq b \leq 60$; $\min_{3 \leq b \leq 60} M(D_b) = M(D_3) = 1.32471795724$, which agrees with θ_0 to 43 digits (indeed $D_3^*(-t) = -(t^3 - t - 1)$, so $M(D_3) = M(t^3 - t - 1) = \theta_0$ exactly); $\max_{3 \leq b \leq 60} M(D_b) = M(D_5) = 1.40987172083$; and $M(D_b) \approx 1.3815$ for b near 60.

The brief I was working from tabulated the *maximum single root modulus* of D_b , which decreases towards 1 (1.151 at $b = 3$, 1.048 at $b = 60$), and that reads like the obstruction weakening. It is not. The number of roots off the unit circle is b or $b - 2$, i.e. it grows, and the Mahler measure — which is the *product* over all of them — stays in $[1.3247, 1.4099]$:

b	$\max_\alpha \alpha $	$\#\{\alpha : \alpha \neq 1\}$	$M(D_b)$
3	1.1509639	3	1.324717957
5	1.1873802	5	1.409871721
9	1.1580869	9	1.377364327
15	1.1211157	15	1.380116208
25	1.0881121	25	1.381067302
40	1.0639792	40	1.381612802
60	1.0478416	60	1.381477502

Name the set the number counts: “max root modulus” counts one root, and the quantity in the theorem counts all of them. The two readings diverge in opposite directions.

9 Verification

All computation is pure Python + SymPy + mpmath (SageMath is not installed in this container). Scripts: `projects/code-1009c2-q404/{hl,kf,exact,verify2,roots,diag,nuchek,nuchek2}.py`.

9.1 An independent Green-polynomial engine

Theorem C is an input, so it is re-verified here by an engine built from the *definition*, sharing no code path with the 10-07 derivation (which went through the h -expansion and the coefficients $c_{\lambda\nu}$). The engine constructs P_λ by Gram–Schmidt for $\langle p_\lambda, p_\mu \rangle_t = \delta_{\lambda\mu} z_\lambda \prod_i (1 - t^{\lambda_i})^{-1}$ along a linear extension of dominance, which characterises P_λ uniquely; then the full Green matrix is a single inverse, $Y = A^{-1}$ with the columns of A the P_μ in the p -basis.

- **Instrument check.** m_λ in the p -basis, verified by evaluating both sides in n variables at random rationals: 44 partitions, $n \leq 7$, 0 failures. (This caught two successive convention errors of mine in the Kostka relation.)

- $P_{(2)} = m_2 + (1-t)m_{11}$, $P_{(3)} = m_3 + (1-t)m_{21} + (1-t)^2m_{111}$, $P_{(2,1)} = m_{21} + (2+t)(1-t)m_{111}$, $P_{(1^n)} = m_{1^n}$ — all as published.
- **Check 1** ($t = 0$, independent mechanism): $Y_\rho^\mu(0) = \chi_\rho^\mu$, against a Murnaghan–Nakayama character routine. 434 pairs, $n \leq 7$, 0 failures. Positive control $Y \mapsto Y + 1$: fires on 434 of 434.
- **Check 2** ($t = 1$, independent mechanism): $Y_\rho^\mu(1) = [m_\mu]p_\rho$, against a direct count of functions from the parts of ρ to the rows of μ with prescribed fibre sums. 434 pairs, $n \leq 7$, 0 failures.
- **Check 3** (Theorem C): the closed form (1) reproduced on all 28 two-row $\times$ two-part cells with $n \leq 7$, 0 failures. Positive control: dropping the $\llbracket a = b \rrbracket$ term was predicted in advance to fire on the 3 cells with $a = b$ and $m = b$, and fired on exactly that set.

9.2 The classification and the obstruction

- **Exact (non-numerical) cyclotomicity test.** “Monomial times a product of cyclotomics” is decided by rational factorisation plus matching each monic irreducible factor against Φ_e for all e with $\varphi(e) = \deg$. Self-tested in both arms: it accepts t , Φ_6 , $t^6 - 1$, $1 - t^{12}$, $(t - 1)t^4(t^3 + 1)$, $2t$, and rejects $t^3 - t - 1$ and $t^2 - t + 2$.
- D_b is a monomial times a product of cyclotomics exactly for $b \leq 2$ and never for $3 \leq b \leq 40$: 0 mismatches against Theorem 3.3.
- E_b exactly for $b = 1$ and never for $2 \leq b \leq 40$: 0 mismatches against Theorem 4.1.
- Lemma 3.1, two ways: numerically, the roots of D_b on S^1 are exactly $\{e^{\pm i\pi/3}\}$ when $b \equiv 2 \pmod{6}$ and none otherwise ($b \leq 40$, 0 failures); exactly, $\Phi_6 \mid D_b \iff b \equiv 2 \pmod{6}$ by polynomial division ($b \leq 40$, 0 failures), the divisor set being $\{2, 8, 14, 20, 26, 32, 38\}$. Lemma 3.2: $\gcd(D_b, D'_b)$ has degree 0 for all $b \leq 40$.
- Theorem 4.1: $E_b(0) = 2$ for $2 \leq b \leq 50$; the minimum root modulus is ≥ 1 with the smallest observed value exactly 1.0; $M(E_b) = \nu(E_b) = 2$ to 25 digits, 0 failures.
- Proposition 5.1: all 465 off-diagonal cells with $b \leq 30$ are monomials times products of cyclotomics, 0 failures; and the claimed closed forms agree with (1) on 105 + 15 instances, 0 failures.
- Theorem 6.1: the predicate (vi) matched against the exact factorisation test on all 28 engine-computed cells with $n \leq 7$, 0 failures. The non-unitary cells found were exactly $\{((2, 2), (2, 2)), ((3, 3), (3, 3)), ((4, 3), (4, 3))\}$, i.e. E_2, E_3, D_3 — which is what (vi) predicts.
- Corollary 7.2: ν computed for all four old witnesses — $\nu(t^2 - t + 2) = \nu(t^2 - 2t + 2) = 2$ and $\nu(t^3 - t^2 + 1) = \nu(t^3 + t^2 - 1) = 1.754877666$, equal to 10 digits as the corollary requires — and none of the four has a root on S^1 . The identity $\alpha^3 + \alpha^2 - 1 = \alpha^2$ at $\alpha = e^{2i\pi/3}$ checked to 25 digits.
- Proposition 8.1: $D_b^* - D_b = t^{b-1} - t$ verified symbolically for $1 \leq b \leq 60$, 0 failures; $D_b^* \neq \pm D_b$ for $3 \leq b \leq 60$ and $D_2^* = D_2$.

9.3 Two mechanisms for M , and what the disagreement was

$M(D_b)$ was computed (i) from high-precision roots and (ii) from Jensen’s formula $M(f) = \exp \int_0^1 \log |f(e^{2\pi i \theta})| d\theta$ — genuinely different mechanisms. The first comparison reported 36 of 40 “failures” at tolerance 10^{-12} . That was the *instrument*, not the value: when $\Phi_6 \mid D_b$ the integrand has a logarithmic singularity on the contour, and even otherwise roots approach S^1 as b grows. After deflating Φ_6 exactly (which changes no value, as $M(\Phi_6) = 1$), the two mechanisms agree to $\leq 1.6 \times 10^{-6}$ over $1 \leq b \leq 40$, while the quadrature’s own convergence estimate $|J_{\deg 8} - J_{\deg 12}|$ is as large as 4.3×10^{-3} — so the residual is quadrature error, and the quadrature is the weaker instrument. On a control with no root near the contour, $t^3 - t - 1$, the two agree to 12 digits; on a deliberately mismatched pair ($M_{\text{roots}}(t^3 - t - 1)$ against $M_{\text{Jensen}}(D_5)$) the discrepancy is 0.085, so the comparison is not vacuous.

Separately, $\nu(D_b) = M(D_b)^2$ was checked for $2 \leq b \leq 60$ with 0 failures at 10^{-25} ; this is Lemma 2.3 and is a consistency identity, not a second measurement.

9.4 A convention error worth recording

My first engine computed Kostka–Foulkes polynomials by the Lascoux–Schützenberger charge. It was wrong twice: first it computed *cocharge* ($K_{\lambda\lambda}$ read $t^{n(\lambda)}$ and $K_{(n),\mu}$ read 1 — exactly swapped), and after the index rule was corrected the asymmetric degree test $\deg K_{\lambda\mu} = n(\mu) - n(\lambda)$ with leading coefficient 1 still failed on 133 of 471 nonzero pairs ($K_{(4,1),(2,2,1)}$ read $2t^3$, leading coefficient 2). The three symmetric tests ($K_{\lambda\lambda} = 1$, $K_{(n),\mu} = t^{n(\mu)}$, $K(1) = \#\text{SSYT}$) all *passed* at that point. I abandoned that route and built the definition-based engine instead, which is why nothing above depends on charge. The lesson is the asymmetric test: the three symmetric ones were invariant under the reading-word convention I had wrong.

10 What this changes, and what it does not

1. **Theorem D’s hypothesis “ b odd ≥ 3 ” can be deleted** in favour of “ $b \geq 3$ ”, and the companion family $a = b \geq 2$ can be added. Registry node **even- b -non-cyclotomicity** (currently **computed**) is now proved, by Lemma 3.1, and the real-root route it was waiting for is not the one that worked.
2. **The statement is strictly stronger than “not a product of cyclotomics”**: it is $\nu > 1$, which also rules out rational scalars and any class of factors with all roots on $S^1 \cup \{0\}$.
3. **Theorem 6.1 is new**: the obstruction’s locus is now known exactly on the two-row slice, with the three exceptions exhibited.
4. **What is *not* claimed**. D_b is not asserted irreducible — it is not; e.g. $D_8 = \Phi_6 \cdot (t^6 - t^4 - t^3 + t + 1)$. Theorem 3.3(d) states exactly what the cyclotomic part is. Nothing here says anything about $\ell(\lambda) \geq 3$, where the fourth witness $t^3 + t^2 - 1$ of the old scope corollary lives. And Remark 8.2 is explicitly conditional on a citation I did not re-read.

Question 10.1 (new). Is $\lim_{b \rightarrow \infty} M(D_b)$ equal to $\exp(m(1 + x + y)) = 1.38136\dots$, Smyth’s two-variable Mahler measure? Writing $D_b = y(t - 1) + 1$ with $y = t^{b-1}$ suggests a Lawton-type limit, and the measured values hover near 1.3815 for $b \in [50, 60]$. This is an observation about my own family, not a claim; both cited theorems (Lawton, Smyth) would need reading.

Question 10.2. Does the two-circles mechanism extend to $\ell(\lambda) \geq 3$ in general? For the one witness I have there it does, in two independent ways: by Corollary 7.2 it is $-D_3(-t)$, and directly — if $|\alpha| = 1$ and $\alpha^3 + \alpha^2 - 1 = 0$ then $\alpha^2(\alpha + 1) = 1$ forces $|\alpha + 1| = 1$, the same intersection of two unit circles but now centred at 0 and -1 , giving $\alpha = e^{\pm 2i\pi/3}$ with $\alpha^3 = 1$, whence $\alpha^3 + \alpha^2 - 1 = \alpha^2 \neq 0$: no root on S^1 at all. What is missing is a closed form for Y_ρ^λ at $\ell(\lambda) = 3$ to apply it to; Theorem C covers only $\ell(\lambda) \leq 2$.